

2920/206
DATABASE MANAGEMENT SYSTEMS
November 2018
Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE II
DATABASE MANAGEMENT SYSTEMS

3 hours

INSTRUCTIONS TO CANDIDATES

*This paper consists of EIGHT questions.
Answer any FIVE questions in the answer booklet provided.
ALL questions carry equal marks.
Candidates should answer the questions in English.*

This paper consists of 5 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

1.
 - (a) Outline **four** characteristics of a Database Management System. (4 marks)
 - (b) Explain the Three-Schema Architecture of a Database Management System. (4 marks)
 - (c)
 - (i) Outline **two** threats to data stored in a Database Management System (2 marks)
 - (ii) A newly established manufacturing company intends to design a database from scratch. State **four** relational database languages they could use. (4 marks)
 - (d) With the aid of an appropriate symbol, explain each of the following components of an Entity- Relationship Diagram (ERD):
 - (i) entity;
 - (ii) relationship;
 - (iii) attribute. (6 marks)
2.
 - (a) Outline **three** duties of a Database Administrator (3 marks)
 - (b) Explain each of the following types of database architectures:
 - (i) Client Server;
 - (ii) Distributed. (4 marks)
 - (c) With the aid of examples in each case, distinguish between Data Definition Language (DDL) and Data Manipulation Language (DML). (6 marks)
 - (d) During a database maintenance session, it was discovered that the Data dictionary had malfunctioned.
 - (i) Explain **two** functions of this dictionary in a Database. (4 marks)
 - (ii) State **three** Components of this dictionary. (3 marks)
3.
 - (a) Outline **three** properties of a relational table. (3 marks)
 - (b) A company intends to integrate its Database Management System. Explain **three** structural components that may be considered. (6 marks)
 - (c) Differentiate between *superkey* and *candidate key* as used in Database management systems. (4 marks)
 - (d)
 - (i) Describe the object oriented database model. (4 marks)
 - (ii) An ICT Technician intends to perform database recovery procedures from the Enterprise's backup. State **three** types of backup that he may use in the recovery process. (3 marks)
4.
 - (a) Define each of the following terms as used in the Entity-Relationship Modelling:
 - (i) specialisation;
 - (ii) aggregation. (4 marks)

- (b) With the aid of an example in each case, distinguish between *composite* and *derived* attributes. (4 marks)
- (c) Explain each of the following terms as used in relational databases:
- Cardinality ratio;
 - Participation constraint. (4 marks)
- (d) The following narrative is a representation of information about banks. Use it to answer the questions that follow
- There are multiple banks and each bank has many branches. Each branch has Multiple customers.
 - Customer has a name, address that consists of house number, area and city, and one or more phone numbers.
 - Customers have various types of accounts.
 - Some Customers also had taken different types of loans from these bank branches.
 - Account has number, type and balance.
 - One customer can have multiple accounts and Loans.
- Identify **four** Entities in the narrative. (2 marks)
 - Draw an Entity-Relationship Diagram to represent the information. (6 marks)
5. (a) Outline **three** characteristics of the First Normal Form (1NF). (3 marks)
- (b) Explain each of the following types of anomalies in Normalization:
- Update;
 - Deletion. (4 marks)

- (c) Table 1 shows information contained in a database table. Use it to answer the questions that follow:

roll_no	Name	Subject
101	John	OS, CN
103	Jeremy	Java
102	Jacob	C, C++

Table 1

Normalize the table up to the second normal form (2NF). (7 marks)

- (d) (i) Peter designed a Database Management System using a relational Model. Outline **three** challenges that he may have encountered during the design. (3 marks)
- (ii) State **three** database schemas he may have used. (3 marks)
- 6/ (a) Outline **four** trends in database technology. (4 marks)
- (b) Explain the referential Integrity rule. (2 marks)
- (c) Peter has been tasked to design a database by following the database design life cycle.
- (i) Explain the conceptual modelling stage of this life cycle. (2 marks)
- (ii) Outline **four** advantages of the model in (i) (4 marks)
- (d) Table 2 and table 3 shows fruits and snacks respectively available at different Kiosks. Use them to answer the questions that follow:

FRUIT

Fruit code	A	B	C
F01	1	2	3
F02	4	5	6
F03	7	8	9

Table 2

SNACK

Snack code	B	C	D
S01	2	3	10
S02	2	3	11
S03	6	7	12

Table 3

Compute

- (i) $\Pi_{A,C}(\text{FRUIT})$;
- (ii) $\sigma_{B=2}(\text{SNACK})$;
- (iii) Natural join;
- (iv) Outer join. (8 marks)
7. / (a) Outline **two** uses of database systems in hotel industry. (2 marks)
- (b) Distinguish between *weak* and *strong* entities. (4 marks)
- (c) A database has been scheduled to run an automatic update on a daily basis. Explain **three** types of integrity constraints that must be checked during update operation. (6 marks)